

PROJECT PROPOSAL

Predicting Patient Satisfaction from Hospital Reviews Using Traditional Machine Learning

IE7275 Data Mining in Engineering | Wiktoria Lasek

Problem Setting

Access to healthcare in Canada remains a significant challenge. Millions of Canadians are unable to see a doctor in a timely manner, and when they do receive care, the quality of their experience varies widely across hospitals and clinics. Patients frequently share their experiences through online reviews, providing hospitals with a valuable but largely underutilized source of feedback. Healthcare administrators and hospital managements teams receive hundreds or thousands of these reviews but lack the time and tools to analyse them systematically at scale (Greaves et al., 2013).

The inability to process patient feedback efficiently means that recurring issues – such as long wait times and poor communication often go unaddressed. Automated tools that can classify and analyse patient sentiment would enable healthcare providers to identify problem areas quickly, prioritize improvements, and ultimately deliver better patient care (Manning et al., 2008).

Problem Definition

This project aims to build an automated system that reads hospital patient reviews and predicts whether the experience was positive or negative. Using traditional machine learning classification techniques, the goal is to identify which model best captures patient sentiment from text — and which words and phrases most strongly indicate satisfaction or dissatisfaction.

The outcome is a practical tool that healthcare providers can use to automatically monitor patient feedback at scale, identify recurring issues, and make data-driven decisions to improve the quality of care.

Data Source and Description

The dataset used in this project is the Hospital Patient Reviews Dataset, publicly available on Kaggle (Hospital Reviews Dataset, 2023, <https://www.kaggle.com/datasets/junaaid6731/hospital-reviews-dataset>).

Dataset name	Hospital Patient Reviews Dataset
Source	Kaggle
Number of Rows	997 patient reviews
Number of Columns	3 columns
Variable names	Feedback, Sentiment label (0,1), Ratings (1-5 stars)
Train/Test Split	80% training, 20% test
Validation Strategy	5-fold cross validation

Methodology

This project follows the full data mining process across the following stages:

- 1. Data exploration and visualization:** The dataset will be explored to understand the overall structure, patterns, and distribution of patient feedback. Visual summaries will be created to communicate key findings.
- 2. Data preprocessing:** The review text will be cleaned and transformed into numerical features using TF-IDF vectorization. Standard text preprocessing techniques will be applied to prepare the data for modelling.

3. **Dimension reduction and feature selection:** Given the high dimensional nature of data, feature selection techniques will be applied to identify the most meaningful and predictive terms.
4. **Model exploration and selection:** Several classification algorithms (Joachims, 1998) will be explored and compared to find the best performing approach for predicting patient satisfaction. Models will be validated using 5-fold cross validation (Manning et al., 2008).
5. **Performance evaluation and visualization:** Models will be evaluated using standard classification metrics. Results will be visualized to clearly communicate model performance and support the selection of the best classifier.

References

- Greaves, F., Ramirez-Cano, D., Millett, C., Darzi, A., & Donaldson, L. (2013). Harnessing the cloud of patient experience: Using social media to detect poor quality healthcare. *BMJ Quality & Safety*, 22(3), 251-255. <https://doi.org/10.1136/bmjqs-2012-001527>
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- Joachims, T. (1998). Text categorization with support vector machines: Learning with many relevant features. *European Conference on Machine Learning*. <https://doi.org/10.1007/BFb0026683>
- Manning, C. D., Raghavan, P., & Schutze, H. (2008). *Introduction to Information Retrieval*. Cambridge University Press. <https://nlp.stanford.edu/IR-book/>